Tightening Torques for the screws at each joint - 17259

Tightening Torques

The values in list below can also be found din the Service manual except for the Tool/Gripper connection

UR3 tightening torque values

CONNECTION	TORQUE	HEAD SIZE
Base m. bracket/ J0 Base	3.0Nm	Torx T20
[J0] Base / J[1] Shoulder	3.0Nm	Torx T20
[J1] Shoulder/ Upper arm	3.0Nm	Torx T20
Upper arm/ [J2] Elbow	1.3Nm	Torx T10
[J2] Elbow/ Lower arm	1.3Nm	Torx T10
Lower arm/ [J3] Wrist 1	1.3Nm	Torx T10
[J3] Wrist 1/ [J4] Wrist 2	1.3Nm	Torx T10
[J4] Wrist 2/ [J5] Wrist 3	1.3Nm	Torx T10
[J5] Wrist 3/ Tool m. bracket	1.3Nm	Torx T10
Tool / Gripper (M6)	10Nm	
Alignment screw	0.4Nm	Torx T10
Blue lid: Base, Shoulder and Elbow	0.4Nm	Torx T10
Blue lid: Wrist 1, Wrist 2 and Wrist 3	0.4Nm	Torx T8

UR5 tightening torque values

CONNECTION	TORQUE	HEAD SIZE
Base m. bracket/ J0 Base	3.0Nm	7 mm
[J0] Base / J[1] Shoulder	3.0Nm	7 mm
[J1] Shoulder/ Upper arm	3.0Nm	7 mm
Upper arm / [J2] Elbow	3.0Nm	7 mm

[J2] Elbow / Lower arm	3.0Nm	7 mm
Lower arm/ [J3] Wrist 1	1.3Nm	5.5 mm
[J3] Wrist 1/ [J4] Wrist 2	1.3Nm	5.5 mm
[J4] Wrist 2/ [J5] Wrist 3	1.3Nm	5.5 mm
[J5] Wrist 3/ Tool m. bracket	1.3Nm	5.5 mm
Tool / Gripper (M6)	9.0Nm	
Alignment screw	0.4Nm	Torx T10
Blue lid	0.4Nm	Torx T10

UR10 tightening torque values

CONNECTION	TORQUE	HEAD SIZE
Base m. brcket/ J0 Base	8.0Nm	10 mm
[J0] Base / J[1] Shoulder	8.0Nm	10 mm
[J1] Shoulder/ Upper arm	8.0Nm	10 mm
Upper arm/ [J2] Elbow	3.0Nm	7 mm
[J2] Elbow/ Elbow Counterpart	3.0Nm	7 mm
Elbow Counterpart / Lower arm	3.0Nm	7 mm
Lower arm/ [J3] Wrist 1	1.3Nm	5.5 mm
[J3] Wrist 1/ [J4] Wrist 2	1.3Nm	5.5 mm
[J4] Wrist 2/ [J5] Wrist 3	1.3Nm	5.5 mm
[J5] Wrist 3/ Tool m. bracket	1.3Nm	5.5 mm
Tool / Gripper (M6)	9.0Nm	
Alignment screw	0.4Nm	Torx T10
Blue lid	0.4Nm	Torx T10